

Mathematics for Computer Science – Group, Subgroup and Cyclic Group

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提醒.

提醒.

以下群论内容不仅仅是开启了新的学习方向，而是对之前数论相关内容的重新审视和抽象，形成一次二重奏。而且，我们会从思考与抽象中得到更多的洞见，还有乐趣！

抽象.

抽象：发现已知的世界中事物的共性与区别，忽略掉某些细节，得到一种更通用、更宽泛、可以描述更广阔世界的框架或语言，从而探求未知世界的知识。

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Question.

更给出（或找出）更适合你自己的“抽象”的定义。

重新审视 “消去律” .

In last chapter, we extremely rely on *Cancellation Law*.

If $\gcd(c, m) = 1$ and $ac \equiv bc \pmod{m}$, then

$$a \equiv b \pmod{m}.$$

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Question.

Actually, what is cancellation?

Ideas.

From $ac \equiv bc \pmod{m}$ to $a \equiv b \pmod{m}$, seemingly, we need division, actually we need multiplication:

$$acc^{-1} \equiv bcc^{-1} \pmod{m}.$$

Hence by $cc^{-1} \equiv 1 \pmod{m}$, we have

$$a \equiv b \pmod{m}.$$

Why c^{-1} exists? Because of $\gcd(c, m) = 1$!

Motivation.

Additional conditions.

Need more conditions?

Motivation.

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Yes, we need *associativity*:

$$(ac)c^{-1} \equiv a(cc^{-1}) \pmod{m},$$

and *closure*, which means that we only consider numbers from a specified set.

Motivation.

Additional conditions.

Need more conditions?

Yes, we need *associativity*:

$$(ac)c^{-1} \equiv a(cc^{-1}) \pmod{m},$$

and *closure*, which means that we only consider numbers from a specified set.

Question.

什么数学操作不满足结合律?

Wrap up.

Wrap these up. For a set $\mathbb{G}$ and an operator $\cdot$ on the elements, we need:

- *Closure*: $\forall a, b \in \mathbb{G}, \quad a \cdot b \in \mathbb{G}.$
- *Associativity*: $(a \cdot b) \cdot c = a \cdot (b \cdot c).$
- An element "1" called *identity*, such that $1 \cdot a = a \cdot 1 = a.$
- $\forall a \in \mathbb{G}$, there exists $a^{-1} \in \mathbb{G}$, such that $a \cdot a^{-1} = 1 = a^{-1}a$, called *inverse*.

Definition of Group.

A group is a set $\mathbb{G}$ together with an operation $\cdot$ satisfying the following axioms:

- *Closure(封闭性)*: $\forall a, b \in \mathbb{G}, \quad a \cdot b \in \mathbb{G}$.
- *Associativity(结合律)*: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- There is an element $e \in \mathbb{G}$, called *identity(单位元)*, such that $e \cdot a = a \cdot e = a$.
- $\forall a \in \mathbb{G}$, there exists $a^{-1} \in \mathbb{G}$, called *inverse(逆元)*, such that $a \cdot a^{-1} = e = a^{-1} \cdot a$.

Examples of groups.

Groups.

- $(\mathbb{Z}, +)$ is a group, while $(\mathbb{Z}, \times)$ is not a group.
- $(\mathbb{Q}, +)$ and $(\mathbb{R}, +)$ are groups.
- $(\mathbb{Q}^*, \times)$ and $(\mathbb{R}^*, \times)$ are groups.

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- $(\mathbb{Q}^*, \times)$ and $(\mathbb{R}^*, \times)$ are groups.

Check.

Please check and know why.

Examples of some important groups.

$\mathbb{Z}_n$

Let n be an integer, the set $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ forms a group under the operation of addition. However, $(\mathbb{Z}_n, \times)$ is not a group.

Examples of some important groups.

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Let n be an integer, the set $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ forms a group under the operation of addition. However, $(\mathbb{Z}_n, \times)$ is not a group.

$$\mathbb{Z}_p^*$$

Let p be a prime number, the set $\mathbb{Z}_p^* = \{1, 2, \dots, p-1\}$ forms a group under the operation of multiplication. (Recall Fermat's Little Theorem.)

Examples of some important groups.

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$$\mathbb{Z}_n^*$$

Let n be an integer, the set $\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}$ forms a group under the operation of multiplication. (Recall Euler's Theorem.)

An important group: n th root of unity.

Example

设 n 为正整数, 称方程 $z^n = 1$ 在复数上的所有解为 n 次单位根, 记为 $\mathbb{U}_n = \{z \in \mathbb{C} : z^n = 1\}$ 。根据高等数学相关知识可知, $\mathbb{U}_n = \{e^{\frac{2\pi ik}{n}}, k = 0, 1, \dots, n-1\}$, 并且

$$e^{\frac{2\pi ik}{n}} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), k = 0, 1, \dots, n-1$$

如果记 $\omega = \cos(\frac{2\pi}{n}) + i\sin(\frac{2\pi}{n})$, 则可更简洁地表示 n 次单位根为

$$\mathbb{U}_n = \{\omega^i, i = 0, 1, \dots, n-1\}$$

可证明 $\mathbb{U}_n$ 在复数的乘法上形成群 (这是课后习题), 并称 $\mathbb{U}_n$ 为 n 次单位根群。

8th root of unity.

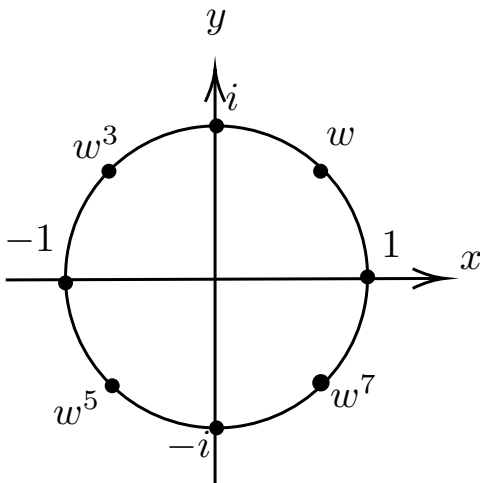


Figure: 8 次单位根群

Thinking.

$\mathbb{Z}_n$ 与 $\mathbb{U}_n$ 的操作行为很类似吗？这会意味着什么呢？另外， $\mathbb{U}_n$ 的操作行为与 $\mathbb{Z}_p^*$ 不很类似，是吗？

Basic Properties of Groups.

Proposition

Proposition 1. The identity element in a group $\mathbb{G}$ is unique; that is, there exists only one element $e \in \mathbb{G}$ such that $eg = ge = g$ for all $g \in \mathbb{G}$.

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Proof.

Suppose $\exists e, e' \in \mathbb{G}$ are identities. Then:

- $ee' = e'$
- $ee' = e$

Combining these two equations, we have $e = ee' = e'$. □

Proposition 2.

Proposition

Proposition 2. If $\forall g \in \mathbb{G}$, then the inverse of g , g^{-1} , is unique.

Proof.

如果 g^{-1} 和 g' 都是 g 的逆元, 则有

$$g' = g'e = g'(gg^{-1}) = (g'g)g^{-1} = eg^{-1} = g^{-1}.$$



Proposition 3.

Proposition

Proposition 3. Let $\mathbb{G}$ be a group. If $a, b \in \mathbb{G}$, then $(ab)^{-1} = b^{-1}a^{-1}$.

Proof.

By construction.

- $ab(b^{-1}a^{-1}) = e.$
- $(b^{-1}a^{-1})ab = e.$

Combining these two equations, we know, the inverse of (ab) is $b^{-1}a^{-1}$. □

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Proof.

By definition, $gg^{-1} = e$ and $g^{-1}(g^{-1})^{-1} = e$. Hence:

$$(g^{-1})^{-1} = (gg^{-1})(g^{-1})^{-1} = g(g^{-1}(g^{-1})^{-1}) = ge = g.$$



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另一种思路.

该命题要证明的是 g^{-1} 的逆元是 g 。根据 g^{-1} 的定义，“交换” g 与 g^{-1} 的位置，即得！

Proposition 5.

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Proof.

- Existence. Such an x exists.
- Uniqueness. Suppose that x_1 and x_2 are both solutions ...

Left as an exercise.



Proposition 6.

Proposition

Cancellation Law. Let $\mathbb{G}$ be a group, and $a, b, c \in \mathbb{G}$. Then $ba = ca$ implies $b = c$ and $ab = ac$ implies $b = c$.

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Proposition 6.

Proposition

Cancellation Law. Let $\mathbb{G}$ be a group, and $a, b, c \in \mathbb{G}$. Then $ba = ca$ implies $b = c$ and $ab = ac$ implies $b = c$.

Proof.

Left as an exercise. □

A little thought.

Where does "*Cancellation Law*" come from?

置换与消去律

在费尔马小定理和欧拉定理的证明中，依赖消去律可得：

对任意素数 p 和与 p 互素的正整数 a ,

$$\mathbb{Z}_p^* = a\mathbb{Z}_p^* = \{ai : \forall i \in \mathbb{Z}_p^*\};$$

对任意合数 n 和与 n 互素的正整数 a , $a\mathbb{Z}_n^* = \mathbb{Z}_n^*$ 。

请问，对任意的群 $\mathbb{G}$ 和群元 a ，是否有

$\mathbb{G} = a\mathbb{G} = \{ag : \forall g \in \mathbb{G}\}$ ？为什么？

逆元的唯一性源自什么？

回忆，模 n 的乘法逆元源自 Bezout 定理，然后我们分析过为什么乘法逆元唯一。请问，对任意的群 $\mathbb{G}$ ，我们也有“逆元唯一”的结论，此时，逆元唯一源自于什么？这样的思考能给我们什么启发？

课堂练习-1.

设 a 和 b 是群 $\mathbb{G}$ 中的元素。如果 $a^4b = ba$ 且 $a^3 = e$, 证明 $ab = ba$ 。

课堂练习-2.

设 a 和 b 是群 $\mathbb{G}$ 中的元素。证明, 对任意 $n \in \mathbb{Z}$, 有 $ab^n a^{-1} = (aba^{-1})^n$ 。

Notations.

Let $\mathbb{G}$ be a group, and $g \in \mathbb{G}$. For $n \in \mathbb{N}$.

Notations.

- $g^0 = e$
- $g^n = \underbrace{g \cdot g \cdots g}_{n \text{ times}}$
- $g^{-n} = \underbrace{g^{-1} \cdot g^{-1} \cdots g^{-1}}_{n \text{ times}}$

Order.

The order of a finite group is the number of elements that it contains. If $\mathbb{G}$ is a group containing n elements, we write $|\mathbb{G}| = n$.

Definition

(Subgroup.) (子群)

Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subset of $\mathbb{G}$. If $\mathbb{H}$ is a group under group operation in $\mathbb{G}$, then $\mathbb{H}$ is said to be a subgroup of $\mathbb{G}$, denoted by $\mathbb{H} \leq \mathbb{G}$.

Examples of Subgroup.

Examples of Subgroup.

- For any group $\mathbb{G}$, there is a trivial subgroup $\{e\}$.
- The additive groups: $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$.
- $\forall n \in \mathbb{Z}$, $n\mathbb{Z} = \{kn \mid k \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

Examples of Subgroup.

Subgroup of $\mathbb{Z}_p^*$.

Let p be a prime, for all $i \in \mathbb{Z}_p^*$, compute $i^2 \bmod p$, form a set $\mathbb{S} = \{i^2 \bmod p, \forall i \in \mathbb{Z}_p^*\}$. Check that $\mathbb{S}$ is a group under the operation of multiplication, namely $\mathbb{S}$ is a subgroup of $\mathbb{Z}_p^*$. What is the order of $\mathbb{S}$?

Exercise.

Write a program to play with $\mathbb{Z}_n^*$.

- Given an integer n , construct the multiplicative group $\mathbb{Z}_n^*$;
- Find a subgroup of the group $\mathbb{Z}_n^*$;
- Find a relation between the size of subgroup and the size of $\mathbb{Z}_n^*$.

Properties of subgroup.

Proposition

(Subgroup.) A nonempty subset $\mathbb{H}$ of a group $\mathbb{G}$ is a subgroup of $\mathbb{G}$ if and only if $\mathbb{H} \neq \emptyset$, and $ab^{-1} \in \mathbb{H}$ for all $a, b \in \mathbb{H}$.

Proof.

Two directions. The $\rightarrow$ part is easy. For $\leftarrow$ part, you need to check that $\mathbb{H}$ satisfies all the axioms of a group. □

有多少种判定子群的方法？

给定群 G 和其子集 H ，如何判断 H 是否 G 的子群？

- 验证四条群公理.
- 命题 6.8: 封闭性、有逆元.
- 命题 6.9: $H \neq \emptyset$, and $ab^{-1} \in H$ for all $a, b \in H$.
- 命题 6.10: 针对有限群，只需要验证封闭性.

练习题.

- 证明：任意群 G 的两个子群的交集也是群 G 的子群。
- 证明或证伪：任意群 G 的两个子群的并集也是群 G 的子群。
- G 是阿贝尔群， H 和 K 是 G 的子群。请证明 $HK = \{hk : h \in H, k \in K\}$ 是群 G 的子群。如果 G 不是阿贝尔群，结论是否依然成立？
- 设 G 是阿贝尔群， m 是任意整数，记 $G^m = \{g^m : g \in G\}$ 。请证明 G^m 是 G 的一个子群。
- 设 G 是阿贝尔群， m 是任意整数，记 $G[m] = \{g \in G : g^m = e\}$ 。请证明 $G[m]$ 是 G 的一个子群。

证明题.

证明：阿贝尔群 $\mathbb{G}$ 中的有限阶元素形成子群。该子群有一个特殊的名字，称为挠子群 (*Torsion Subgroup*)。

课堂练习.

证明：对任意的 $n \in \mathbb{Z}$, $n\mathbb{Z}$ 是 $\mathbb{Z}$ 的子群, 且 $\mathbb{Z}$ 中的任意子群只能是某个 $n\mathbb{Z}$ 。

Cyclic Groups (循环群) .

Example of cyclic group.

Consider the following computation: Choose a number g from $\mathbb{Z}_p^*$ randomly, p is a prime, and compute:

$$\mathbb{S} = \{g, g^2, g^3, \dots, g^j, \dots\}$$

Cyclic Groups (循环群) .

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Questions:

- May $\mathbb{S}$ be finite?
- May $\mathbb{S}$ be a group? Why or why not?
- Can $\mathbb{S}$ equals $\mathbb{Z}_p^*$?

Example of cyclic group.

For example: For $p = 11$, choose $g = 4$, and compute:

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We will have:

$$\mathbb{S} = \{4, 5, 9, 3, 1\}$$

Certainly, it is finite and it is a group.

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We will have:

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Certainly, it is finite and it is a group. Questions:

- What will we get if $g = 2$?
- What will we get if $g = 3$?

Theorem

Let $\mathbb{G}$ be a group and g be any element in $\mathbb{G}$. Then the set

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{G}$. We call $\langle g \rangle$ the cyclic group generated by g , and g is a generator of the group.

Proof.

Check the axioms.



Examples of Cyclic groups.

Some cyclic groups.

- $(\mathbb{Z}, +)$ is a cyclic group, while 1 is the generator.
- $(\mathbb{Z}_n, +)$ is a cyclic group, while 1 is the generator.
- $\langle i \rangle$ is a cyclic group, while i is the generator.

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Check.

Please check and know why.

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- Z_n^* is *NOT* a cyclic group, while n is composite.

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Check.

Please check and know why.

单位根群.

$\mathbb{U}_n$ 是循环群吗？为什么，或者为什么不？

Primitive Root (原根)

Definition

Let a and n be relatively prime integers with $n > 0$. The order of a modulo n is the smallest exponent $e \geq 1$ such that $a^e \equiv 1 \pmod{n}$. If the order of a modulo n equals the largest possible order modulo n , then a is called a primitive root modulo n .

原根

From the last example, we know the order of 4 modulo 11 is 5, and the order of 2 and 6 modulo 11 is 10. Since the largest possible order modulo 11 is 10, thus 2 and 6 are two primitive roots modulo 11. Using language of group, we may say that $\mathbb{Z}_{11}^*$ is a cyclic group generated by 2 or 6, and 2 and 6 are generators of $\mathbb{Z}_{11}^*$.

Properties of Cyclic Groups.

Proposition

所有的循环群都是阿贝尔群。

Proposition

循环群 $\mathbb{G} = \langle g \rangle$ 的每一个子群都是循环群。

Properties of Cyclic Groups.

Theorem

Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n . Then $g^k = e$ if and only if n divides k .

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Note that, n is the least positive number such that $g^n = e$.

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Proof.

Note that, n is the least positive number such that $g^n = e$.

1. The $\leftarrow$ part is trivial, since $g^k = g^{ns} = e$.
2. The $\rightarrow$ part. Suppose $g^k = e$. By division algorithm, $k = nq + r$, where $0 \leq r < n$. Hence,

$$e = g^k = g^{nq+r} = g^{nq} g^r = g^r.$$

Thus, $r = 0$.



Properties of Cyclic Groups.

Theorem

Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n . If $h = g^k$ then the order of h is n/d , where $d = \gcd(k, n)$.

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Proof.

Let m be the least positive number such that $h^m = g^{km} = e$.

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Proof.

Let m be the least positive number such that $h^m = g^{km} = e$.

1. Then $n \mid km$, equivalently, $(n/d) \mid (k/d)m$.

Properties of Cyclic Groups.

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Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n . If $h = g^k$ then the order of h is n/d , where $d = \gcd(k, n)$.

Proof.

Let m be the least positive number such that $h^m = g^{km} = e$.

1. Then $n \mid km$, equivalently, $(n/d) \mid (k/d)m$.
2. Since $d = \gcd(k, n)$, n/d and k/d are relatively prime. Thus, $(n/d) \mid (k/d)m$ implies $(n/d) \mid m$. The smallest such m is n/d . $\square$

强调!

循环群元素的阶.

n 阶循环群 $\mathbb{G} = \langle g \rangle$ 中的元素 g^k 的阶为:

$$\text{ord}(g^k) = \frac{n}{\gcd(k, n)}$$

通过生成元找生成元

已知 2 是群 $\mathbb{Z}_{11}^*$ 的生成元，群 $\mathbb{Z}_{11}^*$ 的阶是 10， $2^3 = 8 \in \mathbb{Z}_{11}^*$ ，且 $\gcd(3, 10) = 1$ ，所以 8 的阶是 10，即 8 也是一个生成元。5 不是生成元，因为 $5 = 2^4 \bmod 11$ ， $\gcd(4, 10) = 2$ 。请读者自行验证以上结论。以上命题告诉我们，在知道某个元是生成元时，如何找到另一个生成元。

练习题.

请找出群 $\mathbb{Z}_{11}^*$ 的所有生成元。

思考题.

已知 g 的阶为 n , 则 g^s 的阶 k 必然整除 n 。你能立即想到理由吗？如果 n 为素数, 那么任意 g^s 的阶会是什么？

Properties of Cyclic Groups.

Corollary

Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n then there are exactly $\varphi(n)$ generators in $\mathbb{G}$.

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Proof.

There are n elements in $\mathbb{G}$ with the form g^i , for all $i \in \mathbb{Z}_n$. For arbitrary g^i , its order is n/d , where $d = \gcd(i, n)$, then g^i is a generator when $d = 1$ which means i is relatively prime to n . There are $\varphi(n)$ elements in $\mathbb{Z}_n$ that are relatively prime to n , therefore there are $\varphi(n)$ generators in $\mathbb{G}$. □

Properties of Cyclic Groups.

Corollary

Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order p , where p is a prime, then all elements in $\mathbb{G}$ except e are generators.

Proof.

Trivially from Corollary 16.



Properties of Cyclic Groups.

Corollary

设群 $\mathbb{G} = \langle g \rangle$ 是阶为 n 的循环群, 则任意 $h \in \mathbb{G}$ 的阶必整除 n 。

Proof.

该结论的正确性可直接由命题 7.5 得出。



以下是一种常见的思考方式，请大家注意。

强调！

任意群中阶为 n 的元素 g ，可将群元素 g 的阶视为循环子群 $\langle g \rangle$ 的阶：

$$\text{ord}(g) = |\langle g \rangle| = n.$$

Exercise-1.

- 请心算列举出群 $\mathbb{Z}_{10}$ 的所有生成元。
- 群 $\mathbb{Z}_{17}^*$ 有多少个生成元？已知 3 是其中一个生成元，请问 9 和 10 是否生成元？
- p 和 q 是两个不同的素数，请问 $\mathbb{Z}_{pq}$ 有多少个生成元？ r 是任意正整数，请问 $\mathbb{Z}_{p^r}$ 有多少个生成元？
- 请问 $\mathbb{U}_n$ 中有多少个生成元，它们分别具有什么形式？
- 证明：如果 p 是素数，则 $\mathbb{Z}_p$ 没有非平凡子群。

Exercise.

- 证明：如果群 $\mathbb{G}$ 是阶为 m 的循环群，且 $d \mid m$ ，则 $\mathbb{G}$ 必有 d 阶子群。
- 证明：设 n 为正整数，对任意阶为 n 的循环群 $\mathbb{G}$ ，如果存在整数 d 是 n 的因子，则群 $\mathbb{G}$ 中存在 d 阶元素，且有 $\varphi(d)$ 个 d 阶元素。
- 设 n 是正整数，加法群 $\mathbb{Z}_n$ 是一个循环群。你能否利用本章的知识来证明第四章最后一道习题呢？即： $\sum_{d \mid n} (\varphi(d)) = n$ 。

Properties of Cyclic Groups.

Exercise.

- 设 a 是任意群 $\mathbb{G}$ 的元素。请问子群 $\langle a^m \rangle \cap \langle b^n \rangle$ 的生成元是什么？

Primitive Root Theorem

Theorem

(Primitive Root Theorem.) Every prime p has a primitive root modulo p , and there are exactly $\varphi(p - 1)$ primitive roots modulo p .

General Primitive Root Theorem

Theorem

The group $\mathbb{Z}_n^$ is cyclic if and only if $n = 2, 4, p^e$, or $2p^e$, where p is an odd prime and e is a positive integer.*

原根判定算法

目标与思路.

- 目标：输入 a 和 p ，判断 a 是模 p 的原根
- 算法基础：命题 7.5 展示，任意元素 a 的阶都是 $p-1$ （群阶）的因子；反之，只要 a 的阶不是小于 $p-1$ 的因子，那么 a 的阶就是 $p-1$ ，即为原根。
- 思路：排除法，对群阶的所有因子 d 进行判断，如果有 $a^d \equiv 1 \pmod{p}$ ，则 a 不是原根。
- 实现过程：找出群阶的所有素因子，然后...

原根判定算法

Listing 1: 原根判定算法

```
1 # 输入素数 $p$ 和一个小于 $p$ 的正整数 $a$ 
2 # 输出 Ture, 如果 $a$ 是模 $p$ 的原根, 否则输出 False
3 def is_primitive_root( $a$ ,  $p$ ):
4     flist = prime_fators_list( $p-1$ ) #求 $p-1$ 的所有素因子
5     for  $f$  in flist:
6         if  $\text{pow}(a, (p-1)//f, p) == 1$ :
7             return False
8     return True
```

若干值得注意的问题.

- 该算法的效率该如何衡量？该算法效率的瓶颈在哪里？
- 注意：排除法是枚举所有的因子；但是，算法第一步求的是 $p-1$ 的素因子，不是求出所有因子。
- 算法中求所有素因子的函数如何用 SageMath 方便地实现？

思考.

现在，有群、子群、循环群的概念，试图进一步考察群结构。群结构是目标，子群是研究群结构的基础，构造循环子群是找到子群的最直接方法。那么，我们接下来要做什么？

Definition of Coset.

Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subgroup of $\mathbb{G}$. Define a left coset of $\mathbb{H}$ with representative $g \in \mathbb{G}$ to be the set

$$g\mathbb{H} = \{gh : h \in \mathbb{H}\}.$$

Right coset can be defined similarly by

$$\mathbb{H}g = \{hg : h \in \mathbb{H}\}.$$

Examples of Coset.

Recall our previous proof of Fermat's Little Theorem, we randomly choose a number $a \in \mathbb{Z}_p^*$, and prove

$$a\mathbb{Z}_p^* = \mathbb{Z}_p^*$$

It is similar in Euler's Theorem. $\forall a \in \mathbb{Z}_n^*$,

$$a\mathbb{Z}_n^* = \mathbb{Z}_n^*$$

Examples of Coset.

Let $p = 11$, let $g = 4$, then $\mathbb{H} = \{g^i : i \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}_p^*$.
Actually, $\mathbb{H} = \{1, 3, 4, 5, 9\}$. Compute:

- $\forall a \in \mathbb{H}$, what is $a\mathbb{H}$?
- $\forall a \notin \mathbb{H}$ and $a \in \mathbb{Z}_p^*$, what is $a\mathbb{H}$?

Properties of Coset.

The number of elements in a coset.

Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subgroup of $\mathbb{G}$. $\forall g \in \mathbb{G}$, the number of elements in $\mathbb{H}$ is the same as the number of elements in $g\mathbb{H}$.

Proof.

Define a map $\psi : \mathbb{H} \rightarrow g\mathbb{H}$ by $\psi(h) = gh$. Show that the map is one-to-one (单射) and onto (满射). (Please recall what we have done in the proof of Fermat's Little theorem.) □

Properties of Coset.

Identical or isolation (相等或不相交) .

Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subgroup of $\mathbb{G}$. $\forall g_1, g_2 \in \mathbb{G}$, then $g_1\mathbb{H} = g_2\mathbb{H}$ or $g_1\mathbb{H} \cap g_2\mathbb{H} = \emptyset$.

Properties of Coset.

Identical or isolation (相等或不相交) .

Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subgroup of $\mathbb{G}$. $\forall g_1, g_2 \in \mathbb{G}$, then $g_1\mathbb{H} = g_2\mathbb{H}$ or $g_1\mathbb{H} \cap g_2\mathbb{H} = \emptyset$.

Proof.

Suppose $\exists h_1, h_2 \in \mathbb{H}$ such that $g_1 h_1 = g_2 h_2$, we prove that the $g_1\mathbb{H} \subseteq g_2\mathbb{H}$. Similarly, $g_2\mathbb{H} \subseteq g_1\mathbb{H}$. Then $g_1\mathbb{H} = g_2\mathbb{H}$. Note that:

$$\forall g_1 h \in g_1\mathbb{H}, g_1 h = g_1 (h_1 h_1^{-1}) h = g_2 (h_2 h_1^{-1} h) \in g_2\mathbb{H}$$



Properties of Coset.

Partitioning of group G .

Let G be a group and H a subgroup of G . Then the left cosets of H in G partition G .

Properties of Coset.

Partitioning of group G .

Let G be a group and H a subgroup of G . Then the left cosets of H in G partition G .

Proof.

Nothing! Convince yourself that the cosets gH cover G , and then recall the last proposition. Why cover? Note that $e \in H$! □

Lagrange's Theorem

Notation.

Let G be a finite group and H a subgroup of G . Define the index of H in G to be the number of left cosets of H in G . We denote the index by $[G : H]$.

Lagrange's Theorem (拉格朗日定理) .

Let G be a finite group and H a subgroup of G . Then $|G|/|H| = [G : H]$ is the number of distinct left cosets of H in G .

Proof.

The group G is partitioned in $[G : H]$ distinct left cosets. Each left coset has $|H|$ elements; therefore, $|G| = [G : H]|H|$ □

Corollaries from Lagrange's Theorem

Corollary

Suppose that $\mathbb{G}$ is a finite group and $g \in \mathbb{G}$. Then the order of g must divide $|\mathbb{G}|$.

Corollary

Let $\mathbb{G}$ be a group and $|\mathbb{G}| = p$ where p is a prime. Then $\mathbb{G}$ is cyclic and any $g \in \mathbb{G}$ such that $g \neq e$ is a generator.

Corollary

Let $\mathbb{H}$ and $\mathbb{K}$ be subgroups of a finite group $\mathbb{G}$ such that $\mathbb{K} \subset \mathbb{H} \subset \mathbb{G}$. Then

$$[\mathbb{G} : \mathbb{K}] = [\mathbb{G} : \mathbb{H}][\mathbb{H} : \mathbb{K}]$$

Corollaries from Lagrange's Theorem

Corollary

Fermat's Little Theorem.

$$a^{p-1} \equiv 1 \pmod{p}.$$

Corollary

Euler's Theorem.

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Abstract Fermat's Little Theorem.

Theorem

(Abstract Fermat's Little Theorem.) Let $\mathbb{G}$ be a finite group with order n . Then for any $a \in \mathbb{G}$, $a^n = e$.

(练习题.)

- 设 G 是群, H 是 G 的子群。任取 $g_1, g_2 \in G$, 则 $g_1H = g_2H$ 当且仅当 $g_1^{-1}g_2 \in H$ 。
- 如果 G 是群, H 是群 G 的子群, 且 $[G : H] = 2$, 请证明对任意的 $g \in G$, $gH = Hg$ 。
- 如果 G 是有限群, H 和 K 是群 G 的子群, 且子群 H 的阶与子群 K 的阶互素。请证明, H 与 K 只有一个共同元素-单位元。
- 设 G 是阶为 pq 的群, 其中 p 和 q 是素数。请证明 G 的任意非平凡子群是循环群。
- 设 G 是阶为 pq 的阿贝尔群, 其中 p 和 q 是素数。已知群元 g_1 的阶为 p , 群元 g_2 的阶为 q , 请问群元 g_1g_2 的阶为多少? 请说明理由。